

CHAPTER

Safety

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WARNING

Read this entire chapter before installing, operating or servicing the SalinBloc™ unit. No unit-level hazard analysis, PPE schedule or lockout/tagout procedure specific to SalinBloc™/SonixED™ exists yet in HydroVolta's engineering library — only generic contractor-safety material (tool safety sheets, an electrical risk assessment template, a VCA contractor-competency handbook) and the Safety Data Sheets for the dosing/CIP chemicals. The hazards, PPE and isolation guidance below are derived from those chemical SDSs and from the electrical/pressure/DC figures documented elsewhere in this manual (chapters 10 and 14); they are a defensible starting point, **not** a substitute for a formal hazard analysis, which HydroVolta engineering/EHS must still produce and sign off before this manual is issued to a customer.

2.1 Intended use

The SalinBloc™ unit is intended for the electrodialysis desalination or ion reduction of brackish groundwater, industrial process water, or brine, in the total dissolved solids range HydroVolta publishes as the technology's operating envelope — approximately 1,500–20,000 ppm TDS (see section 3.2) — and within the feedwater quality limits stated for the pretreatment and stack stages (chapters 8 and 10). It is not intended for seawater desalination (HydroVolta states seawater, >35,000 ppm, as outside SonixED™'s primary use case) and is **not** currently certified as a drinking-water treatment system: the 03_UK DWI reg 31 approval folder in HydroVolta's datasheet library contains only generic third-party guidance on how Reg 31 approval works (not an actual approval or application), and the 04_NSF folder contains only a general NSF/ANSI 60/61/372 adoption survey (not an NSF listing) — confirmed by direct inspection of both folders. Do not make a potable-water regulatory claim to a customer until formal approval is obtained — see appendix C. Foreseeable misuse includes: feeding water outside the documented turbidity/iron/chlorine/pH envelope (chapter 10) without adequate pretreatment; bypassing or defeating interlocks, alarms or the emergency stop; operating with the cabinet doors open; and mixing CIP chemicals outside the sequence in chapter 12 (in particular, never combine NaOH with feed/process water — see the warning in chapter 12).

2.2 Qualified personnel

Work on the control cabinet, VFDs and any live electrical circuit, including the SonixED™ stack's DC bus (up to 600 VDC, chapter 10), must only be performed by an electrically qualified person as defined in IEC 60204-1. Chemical dosing tank refilling and CIP chemical handling must only be performed by personnel trained on the relevant Safety Data Sheet for that chemical (section 2.4). HydroVolta's own engineering library uses the Belgian/Dutch **VCA** (Veiligheid, Gezondheid en Milieu Checklist Aannemers) contractor-safety certification as its baseline competency reference for site work; treat VCA-certified (or locally equivalent) status as the expected minimum for any contractor performing installation or maintenance work on the unit. General operators (start/stop, HMI monitoring, routine visual checks) do not need electrical or chemical-handling qualifications but must be trained on this manual and on the local isolation procedure (section 2.7) before working unsupervised.

2.3 Personal protective equipment

- **Chemical dosing / CIP work** (antiscalant, HCl, NaOH, NaOCl, H₂SO₄ — all corrosive per their SDS, section 2.4): chemical-resistant gloves, safety glasses/goggles with a face shield for tank filling or CIP setup, and a chemical-resistant apron. Have an eyewash/emergency shower available wherever CIP or dosing-tank work is performed.
- **Control cabinet / electrical work**: PPE appropriate to the task and the incoming supply voltage (3×400 V AC, chapter 14) per the site's local electrical safety regulation and the qualified person's own risk assessment; always verify isolation (section 2.7) before opening the cabinet.
- **General site / mechanical work** (pumps, valves, piping): safety shoes and, where local site rules require it, hi-vis clothing and hearing protection near running pumps.

2.4 Main hazards

- **Electrical:** the incoming 3×400 V AC cabinet supply (chapter 14), and — specific to this technology — the SonixED™ stack's own DC bus, which is energized to **60–600 V DC at 3–20 A** during normal operation (chapter 10). This DC bus is a hazard independent of the AC supply and must be isolated and verified dead separately before any stack or manifold work.
- **Chemical:** the antiscalant, HCl, NaOH, NaOCl and H₂SO₄ used for dosing and CIP (chapter 12) are corrosive per their Safety Data Sheets (Carl Roth SDS on file in HydroVolta's engineering library). **Never mix NaOH with feed or process water** — HydroVolta's own CIP procedure documents this as a scaling hazard, and mixing acid and hypochlorite streams carries a toxic-gas risk common to all such systems; keep chemical dosing lines and procedures strictly separated as described in chapter 12.
- **Pressurized piping:** the hydraulic circuit operates up to 2.5 bar (stack maximum 3 bar, chapter 10) and the pretreatment UF stage's CEB/backwash cycle briefly reaches similar pressures (chapter 8); depressurize before opening any fitting.
- **Moving parts:** process pumps, dosing pumps, and automated valve actuators (chapters 9 and 11) start automatically under PLC control without warning — treat every pump and valve as capable of starting at any time unless the unit is isolated.

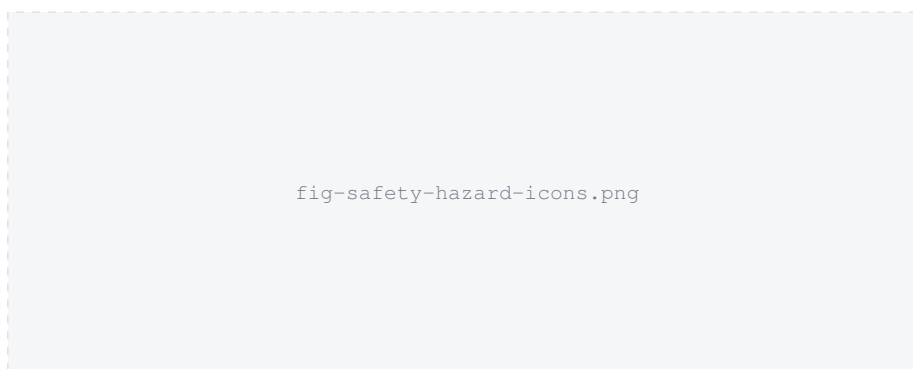


Figure 2.1. Suggested figure: a labeled diagram of the container's exterior and interior showing each hazard zone above (electrical cabinet / DC stack bus, chemical dosing area, pressurized piping, pump/valve zones) with its corresponding ISO 7010 warning pictogram.

2.5 First aid and chemical spill response

The following is transcribed directly from the manufacturer Safety Data Sheets on file for the dosing/CIP chemicals (Carl Roth SDS, Section 4 “First aid measures” and Section 6 “Accidental release measures”; text is consistent across NaOH, HCl, NaOCl and H₂SO₄). Post this section, or an equivalent quick-reference card, wherever chemicals are handled.



Eye contact: flush immediately with plenty of flowing water for **10 to 15 minutes**, holding eyelids apart, and consult an ophthalmologist. Protect the uninjured eye. **Skin contact:** take off all contaminated clothing immediately; wash the affected skin immediately with plenty of water (for H₂SO₄: first mechanically remove the chemical, e.g. dab away with wadding, then wash with a mild cleansing agent and water); seek immediate medical treatment — corrosive injuries left untreated are hard to cure. **Inhalation:** move to fresh air; seek medical advice if symptoms persist or in any case of doubt. **Ingestion:** rinse the mouth immediately and drink plenty of water; call a physician immediately — risk of gastric/esophageal perforation from the corrosive effect.

Spill response (non-emergency personnel only; do not attempt without the PPE in section 2.3): put on PPE, avoid contact with skin, eyes and clothing, and do not breathe vapour/spray. Cover nearby drains to protect surface/ground water — for HCl and H₂SO₄ specifically, the product must be neutralized before any discharge to a drain or sewage plant. Absorb the spill with a liquid-binding material (sand, diatomaceous earth, or an acid/universal binding agent), then place the absorbed material in an appropriate container for disposal. Ventilate the affected area, especially after an HCl release.

2.6 Safety devices and emergency stop

HydroVolta's HMI design specification (chapter 6) confirms a **hardware Emergency Stop button that bypasses the PLC entirely** — see section 6.9 for the full recovery procedure. TODO: its physical location on the control panel/cabinet, and whether more than one E-stop station exists (e.g. one per container, or one per access point), was not confirmed in the documents reviewed — verify against the as-built electrical schematic and IEC 60204-1 on the actual unit before publication. No document describing pressure-relief devices or leak detection was found.

2.7 Isolating the unit before work

No HydroVolta-authored lockout/tagout procedure was found; until one exists, apply this general sequence before any maintenance, sensor, or piping work:

1. Stop the process via the normal `Stop` sequence (chapter 6).
2. Isolate and lock out the incoming electrical supply at the main breaker; verify absence of voltage at the cabinet terminals before touching any wiring.
3. Separately verify the SonixED™ stack's DC bus is de-energized — it operates independently of the AC supply isolation point and can present up to 600 V DC (chapter 10); never assume it is safe just because the AC breaker is open.
4. Isolate and depressurize the hydraulic circuit (close feed/product/ drain isolation points, section 4.4) before opening any fitting.
5. Where chemical dosing lines will be disturbed, close the relevant tank isolation valve and follow the PPE guidance in section 2.3.

TODO: this sequence is assembled from the general electrical/hydraulic/DC facts documented elsewhere in this manual, not from a validated HydroVolta LOTO procedure — HydroVolta engineering/EHS must review and formalize it, including lock/tag hardware and a permit-to-work step if required by the customer's site rules.

2.8 Safety labels on the container

A folder named `17_Photos/Container stickers` exists in HydroVolta’s engineering library, but on inspection it contains only **brand artwork** (the Hydro-Blue and Volta-Yellow logo marks, the HydroVolta wordmark, a “water desalination” decal, and a door logo) — **no hazard/warning pictogram stickers were found there or anywhere else in the library**. This is a real gap: before a unit ships, it needs a proper ISO 7010 hazard-label set (electrical, DC stack, chemical, pressure — matching section 2.4) actually affixed and photographed, and this section rewritten from those photographs.



Figure 2.2. Suggested figure: photograph(s) of the container exterior and control cabinet with each safety/identification label numbered, plus a key table listing what each label means and where it is mounted — to be produced once a real hazard-label set has been designed and affixed to a unit (none currently exists in HydroVolta’s documentation).

